AI WRITING / AI-DETECTION REVIEW REPORT

Manual Assessment of Writing Style and Originality Indicators

Document Under Review

Report Title	Approximation Methodologies for the Travelling Salesman Problem: A Combinatorial Optimization Survey
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Institution	Marwadi University, Department of Computer Science and Engineering – Artificial Intelligence and Machine Learning (CSE – AI/ML)
Course / Guide	Design and Analysis of Algorithms (DAA) – Project Report | Guide: Mr. Sandeep Panchal
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Document Length	40 pages (front matter, nine chapters, references and appendices)
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1. Purpose of This Assessment

This document provides a manual assessment of the writing style of the submitted project report. The goal is to identify passages that exhibit characteristics sometimes associated with AI-assisted writing (for example, highly uniform sentence patterns, generic explanatory phrasing, or repetitive transitional structures) and to contrast them with sections that show project-specific technical detail, experimental design choices, and individual analysis. The assessment is intended to support academic integrity review; it is not a formal detector certificate.

2. Assessment Methodology

The full report (abstract, Chapters 1–9, tables, figure captions, references and appendices) was read manually. Attention was given to sentence-length variation, presence of concrete experimental numbers, explicit statements of assumptions and limitations, citation integration, and the degree of generic versus project-specific language. Targeted attention was also paid to the Abstract, the experimental Chapter 7, the comparative tables, and the conclusion.

Tool status: *No recognised commercial or open AI-detection tool (GPTZero, Originality.ai, Turnitin AI detector, Copyleaks, ZeroGPT, or similar) was accessible for this review. Consequently this document is labelled a* ***Manual AI-Writing Assessment****. No AI-percentage score is reported and no detector output is claimed.*

3. Section-wise Observations

3.1 Front Matter (Title page, Certificate, Acknowledgement, Abstract – approx. pages i–iii)

The title page, certificate and acknowledgement follow conventional institutional templates and show no unusual stylistic markers. The Abstract is dense and information-rich: it states the problem, lists the method families surveyed, names the classical ratios (2 and 3/2), and then reports concrete experimental outcomes (farthest insertion as best constructive method; mean excess reduced from 34.6 % to 5.2 %; bounds held on 300 metric instances but violated on non-metric matrices). These numerical and comparative claims are project-specific and reduce the likelihood that the Abstract is purely generic generated text.

3.2 Chapter 1 – Introduction (approx. pages 1–2)

The background, history and motivation sections use clear but relatively formal academic prose. Some sentences follow a predictable “statement + citation + implication” pattern that is common in both careful human writing and AI-assisted drafting. The problem statement, objectives, scope/limitations and research methodology are written in a direct, list-oriented style that reads as student-authored planning text rather than fluent generated narrative. The explicit limitation that no neural method was implemented is a concrete authorial decision.

3.3 Chapter 2 – Mathematical Foundations (approx. pages 3–4)

Definitions of graphs, cost matrices, symmetric/asymmetric/metric variants, the permutation and ILP formulations, and complexity statements are standard textbook material. The language is precise and somewhat uniform, which is typical when restating classical definitions. The table that distinguishes “optimal tour / proven approximation guarantee / good tour empirically” is a useful conceptual contribution that shows the authors’ organising intent.

3.4 Chapter 3 – Literature Review (approx. pages 5–7)

The chronological organisation (1954–1963, 1970–1978, 1973–2002, etc.) and the two comparison tables are structured and dense. Some paragraphs summarise papers in a balanced, slightly generic style ("X proposed \ldots, Y improved \ldots"). This is normal for a survey; the presence of specific ratios, year ranges and named algorithms (LKH, PTAS, strongly Rayleigh sampling) indicates engagement with the actual literature rather than pure surface paraphrase.

3.5 Chapters 4–6 – Classification, Methodologies, Comparative Analysis (approx. pages 8–18)

Chapter 5 (the longest technical chapter) contains pseudocode, complexity statements, proofs of the 2- and 3/2-approximations, and worked numerical checks on the demonstration instance. These elements—especially the explicit assumptions under which each guarantee holds and the numerical verification of the MST and matching bounds—are characteristic of careful technical writing rather than undifferentiated generated prose. The comparative tables in Chapter 6 are organised by guarantee, assumptions, strengths and limitations; the repeated emphasis that local-search and metaheuristic methods carry no constant-factor guarantee is a consistent authorial theme.

3.6 Chapter 7 – Demonstration and Experimental Study (approx. pages 19–25)

This chapter is the strongest evidence of project-specific work. It describes a single Python file, input validation, independent cost recomputation, an 8-city Manhattan-distance demonstration instance with optimal cost 42, tables of mean gaps and run times, scaling behaviour up to 200 cities, and a controlled 300-instance metric versus non-metric test. Concrete figures (34.6 %, 5.2 %, tree weight 31, matching weight 15, etc.) and the explicit validation protocol are difficult to attribute to generic AI output. The language here is more procedural and less polished than the earlier survey chapters, which is consistent with student experimental reporting.

3.7 Chapters 8–9, References and Appendices

The applications and limitations discussion is brief and measured. The conclusion restates the main supported claims without introducing new unsupported assertions. The reference list contains the expected foundational and recent landmark papers. Appendices supply additional pseudocode, validation notes and sample outputs—further indicators of an implemented system rather than purely textual generation.

4. Writing-Style Analysis

Feature	Observation in the Report	Interpretation
Sentence-length variation	Survey chapters tend toward medium-length, evenly paced sentences; experimental sections contain shorter procedural sentences and longer explanatory ones	Normal academic mixture
Repetitive stock phrases	Occasional reuse of phrases such as "under the triangle inequality", "no constant-factor guarantee", "metric symmetric costs"	Expected technical repetition; not excessive
Generic explanatory tone	Present in background and literature-summary paragraphs; reduced in proofs, pseudocode and experimental discussion	Typical of survey + experiment structure
Project-specific numbers & claims	Abundant in Abstract and Chapter 7 (percentages, instance sizes, demonstration costs, validation protocol)	Strong indicator of original work
Explicit limitations & assumptions	Repeatedly stated (metric assumption required for ratios; no neural implementation; pure-Python scale limit)	Good academic practice; less common in pure AI drafts
Citation integration	Numbered citations appear throughout; key theorems are attributed	Consistent with careful survey writing

5. Repetition and Consistency Observations

The report deliberately repeats the distinction between a proven approximation ratio and an observed empirical gap, and repeatedly reminds the reader that the 2- and 3/2-guarantees require the triangle inequality and symmetry. This thematic consistency is a strength of the argument rather than a sign of unthinking generation. Internal cross-references (to Chapter 7 experiments, to Lemma 5.1, to the demonstration instance) are coherent. No large copy-pasted blocks of identical text across distant sections were observed.

6. Suggestions for Improving Natural Academic Voice (Advisory Only)

The following suggestions are offered solely as advice for future writing. They do not constitute edits to the present report, which remains unchanged.

- In survey paragraphs, occasionally vary sentence openings and length more deliberately so that the rhythm is less evenly paced.
- Where a classical algorithm is restated, a short personal remark on why the assumption matters for the later experiments can further personalise the exposition.

- The experimental chapter already contains the strongest individual voice; extending a similar concrete, slightly less formal tone into the comparative discussion can increase overall naturalness.
- Avoid over-use of the same transitional formula (“This shows that\ldots”, “It is important to note that\ldots”) in consecutive paragraphs.

7. Overall Assessment and Limitations

The report combines a conventional literature survey of a well-studied topic with a concrete, validated experimental component. The survey chapters exhibit the formal, evenly structured prose that is common in both careful student writing and AI-assisted drafting of technical summaries. In contrast, the Abstract, the proofs with numerical checks, the pseudocode, and especially Chapter 7 contain project-specific data, design decisions and validation steps that are characteristic of original student work. No evidence of wholesale unedited AI generation of the experimental content was observed in this manual review.

Limitations of this assessment: (1) it is manual and qualitative; (2) no commercial AI-detection tool was used; (3) writing-style indicators are probabilistic and cannot prove or disprove authorship; (4) the presence of standard textbook material is expected in a survey and should not by itself be treated as an AI indicator. Institutions that require an official AI-detector score should obtain one through their licensed institutional system.

This document is a separate Manual AI-Writing Assessment only. The original project report has not been modified.